

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and executed system validation plans for automotive camera systems, achieving **98% test coverage** against functional and performance requirements.
- Designed test specifications and test methods for mechatronic ECU systems, ensuring alignment with system requirements and standards, and contributing to a **15% reduction in critical defects** found post-release.
- Implemented test automation capabilities using **Python** for automotive software features, increasing regression test throughput by **40%** and enabling earlier detection of integration issues.
- Performed detailed trace analysis and troubleshooting of system issues, validating fixes and reducing root cause analysis time by **20%** through systematic debugging.
- Supported the review of system requirements and architecture designs to assess testability, collaborating with System Engineers to define test coverage and identify potential gaps early in the lifecycle.
- Managed and maintained test environments for bench and HIL setups, ensuring readiness and contributing to the on-time delivery of system validation targets across the project lifecycle.

SKILLS

Programming Languages – Python, C++, C, Java.

Automotive Systems – Advanced Driver Assistance Systems (ADAS), Automotive Camera Systems, Mechatronic ECU Systems, System Validation, Test & Validation Engineering.

Testing Methodologies – System Testing, Functional Requirements, Performance Requirements, Test Specification Design, Test Execution, Trace Analysis, Stress Testing, Robustness Testing, Test Automation.

Development Lifecycle – Systems Development Life Cycle (SDLC), Agile/Scrum Methodologies, CI/CD, Change Management, UAT.

Tools & Technologies – Linux, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace, AWS, Docker, NI LabVIEW & NI TestStand (Awareness).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Automotive ADAS Camera System Validation Suite

- Engineered a comprehensive test suite for ADAS camera systems, integrating **Python** scripting for automated test case execution and results aggregation, leading to a **30% acceleration** in validation cycles.
- Developed detailed test specifications and implemented stress and robustness testing scenarios, achieving **99% requirement coverage** and identifying critical failure points in edge-case performance.

- Collaborated with Systems Engineers to establish traceability between system requirements and test cases within a **Jira**-managed framework, ensuring alignment and facilitating defect tracking for a **10,000+ line** system.

Financial Services Transaction Monitoring Dashboard

- Supported production operations for a financial-services dashboard, implementing enhancements in **Java Spring Boot** to improve transaction-flow monitoring capabilities.
- Developed **Python** scripts for data validation and automated reporting, reducing manual effort in reconciliation workflows by **25%**.
- Managed incident response and SLA adherence for critical payment message processing, maintaining a **99.9% uptime** for operational alerting systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)